N100 Financial Intelligence Platform

Sprint 3 – Screener & Peer Comparison Engine
Bluestock Fintech Internship

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Project Overview

- The N100 Financial Intelligence Platform is a financial analytics solution that processes company financial data stored in SQLite. Sprint 3 focuses on helping analysts identify quality companies through configurable screening criteria and compare companies against sector peers using percentile-based analysis.

Sprint Objectives

- • Build a configurable financial screener.
- • Implement six predefined investment screeners.
- • Calculate a composite quality score.
- • Rank companies within peer groups.
- • Generate Excel reports and radar charts.
- • Validate the complete workflow using automated tests.

Business Problem

- Investors often spend significant time filtering companies manually. This sprint automates the process by applying financial thresholds, generating investment-ready screeners, and benchmarking companies against their peers to support faster decision-making.

System Architecture

- Raw Financial Data → SQLite Database → Pandas Analytics → Screener Engine → Composite Score → Peer Ranking → Excel Reports → Radar Charts.
- The modular architecture separates data storage, analytics, visualization, and reporting.

Day 15 – Filter Engine

- Implemented a configurable filter engine using YAML. The engine loads financial ratios from SQLite, applies threshold-based filtering (ROE, Debt-to-Equity, Free Cash Flow, Market Cap, etc.), and supports analyst-editable configuration without changing Python code.

Day 16 – Preset Screeners

- Developed six investment screeners:
- • Quality Compounder
- • Value Pick
- • Growth Accelerator
- • Dividend Champion
- • Debt Free Bluechip
- • Turnaround Watch

- Each preset applies a different combination of financial metrics to identify companies matching a specific investment strategy.

Day 17 – Composite Quality Score

- Implemented a composite score scaled from 0–100 using profitability, cash flow quality, and operating performance metrics. Companies are ranked in descending order, allowing analysts to prioritize stronger businesses quickly.

Day 18 – Peer Percentile Ranking

- Calculated percentile rankings within peer groups. Metrics such as ROE, Net Profit Margin, Asset Turnover, Debt-to-Equity, and Interest Coverage are compared across companies. Lower Debt-to-Equity receives a higher percentile because lower leverage is preferable.

Day 19 – Radar Charts

- Generated radar charts to visualize company performance across multiple financial dimensions. These charts help compare strengths and weaknesses against sector averages in a single graphical view.

Day 20 – Excel Reporting

- Automatically generated peer_comparison.xlsx and screener_output.xlsx. Reports organize results into sheets, apply conditional formatting, and provide an easy-to-read summary for analysts and business users.

Testing & Validation

- Performed extensive validation of the screener engine, peer ranking logic, and reporting modules. Automated testing with Pytest confirmed that all implemented functionality worked correctly. Final result: 34/34 tests passed.

Technology Stack

- Python 3.11
- Pandas
- SQLite
- OpenPyXL
- Matplotlib
- PyYAML
- Pytest
- Git & GitHub

- These technologies enabled data processing, visualization, reporting, testing, and version control.

Challenges & Solutions

- Challenges included merge-key mismatches, inconsistent year formats, module import issues, Excel formatting, and database validation. Each issue was resolved through debugging, datatype standardization, and improved test coverage.

Key Deliverables

- • Screener Engine
- • Peer Ranking Engine
- • Composite Quality Score
- • Radar Charts
- • Excel Reports
- • SQLite peer_percentiles table
- • YAML Configuration
- • Automated Unit Tests

Learning Outcomes

- Improved skills in data engineering, financial analytics, SQLite database management, Python debugging, report automation, visualization, unit testing, and Git version control.

Conclusion

- Sprint 3 successfully delivered a complete financial screening and peer comparison workflow. The project now supports automated company screening, peer benchmarking, report generation, and validated analytics suitable for business decision support.

Thank You

- Thank you for your time.
- Questions and Discussion